Practical 10:

Aim: Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code:

```
~
  GNU nano 8.7
#include <stdio.h>
#include <stdlib.h>

int main() {
    int requests[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;
    int total = 0;

    int visited[8] = {0};

    printf("Seek Sequence: %d", head);

    for (int i = 0; i < n; i++) {
        int min = 9999;
        int index = -1;

        for (int j = 0; j < n; j++) {
            if (visited[j] == 0) {
                int diff = abs(head - requests[j]);

                // Tie-breaking condition added
                if (diff < min || (diff == min && requests[j] > head)) {
                    min = diff;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        total += abs(head - requests[index]);
        head = requests[index];

        printf(" -> %d", head);
    }

    printf("\nTotal Head Movement = %d\n", total);

    return 0;
}
```

Output:

```
MAYUR LAP MSYS ~
$ gcc SSTF.c -o SSTF

MAYUR LAP MSYS ~
$ ./SSTF
Seek Sequence: 53 -> 65 -> 67 -> 41 -> 14 -> 98 -> 122 -> 183
Total Head Movement = 236
```